FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3843 SPECIAL TOPIC IN DATA ENGINEERING

SECTION 01

MS AZURE TUTORIAL

LECTURER: DR. ARYATI BINTI BAKRI

GROUP 5

Student Name	Matric No.
GOE JIE YING	A23CS0224
LAM YOKE YU	A23CS0233
TAN YI YA	A23CS0187

1. Project Overview

The project focuses on the design and implementation of a full data engineering pipeline using Microsoft Azure tools. The objective of this project is to transform the raw data into meaningful insights for a better decision-making process. The pipeline follows a structured process, beginning with data ingestion, followed by data storage, transformations, analysis and finally visualization. Besides, this project exposes students to real-world data engineering tools, providing practical experience with tools used by industry and enhancing their understanding of concepts learned in lecture.

2. Description of the solution

The solution is built as an end-to-end data pipeline that automates the flow of data, from extract on-premises data, load it into Azure Data Factory and Azure Data Lake Storage Gen 2. Then, it performs necessary transformations using Medallion Architecture in Azure Databricks and utilizes Azure Synapse Analytics to make the data analysis process easier. The transformed data will be fed into Power BI in order to build a customized dashboard for the customers. The pipeline build is set to be triggered every day, ensuring that stakeholders will have access to up-to-date data and insight.

3. Technology Used

This section explains the technology used in this project. The technology used involved Microsoft Azure and PowerBI.

3.1 Azure Data Factory

Azure Data Factory is a fully managed, serverless data integration service that allows data preparation, ETL/ELT processes construction, orchestration and pipeline monitoring.

In this project, Azure Data Factory is used to create a pipeline to ingest data from on-premises sources, store it into storage in Azure Data Lake Storage Gen2 and run data preprocessing procedures using Azure Databricks. In short, Azure Data Factory is used for orchestration and it can be adjusted to trigger in different conditions. In our scenario, it is triggered daily to update the data.

3.2 Azure Data Lake Storage Gen2

Azure Data Lake Storage is a service that implements a set of capability that is built on Azure Blob Storage.

In the project scenario, Azure Data Lake Storage Gen2 is used for storage. Three folders are created to store different layers of data, including Bronze, Silver and Gold.

3.3 Azure Databricks

Azure Databricks is a platform that allows data management, data sharing, data warehousing, data governance, database building, data engineering and so on.

In this project, Azure Databricks is used to perform data transformation, moving data from bronze to silver and then silver to gold.

3.4 Azure Synapse

Azure Synapse Analytics is an end-to-end enterprise analytics service that unifies data integration, big data processing, and data warehousing into a single platform.

In the project context, Azure Synapse is used to serve the data using serverless SQL to query the data lake.

3.5 Power BI

Power BI is a platform for self-service and enterprise business intelligence (BI). It is mainly used to produce data visualisation, detailed analytics and interactive dashboards.

Power BI connects to Synapse endpoints to consume the curated data. It is used to build interactive dashboards for data visualisation.

3.6 Cross-cutting functions

Other cross-cutting functions include security and identity management. The technology used includes Key Vault, that allows us to store secrets and Microsoft Entra ID (formerly known as Azure Active Directory) for identity and access management.

4. Data Pipeline Architecture

This section explains the overall architecture of the end-to-end data engineering pipeline developed using Microsoft Azure services.

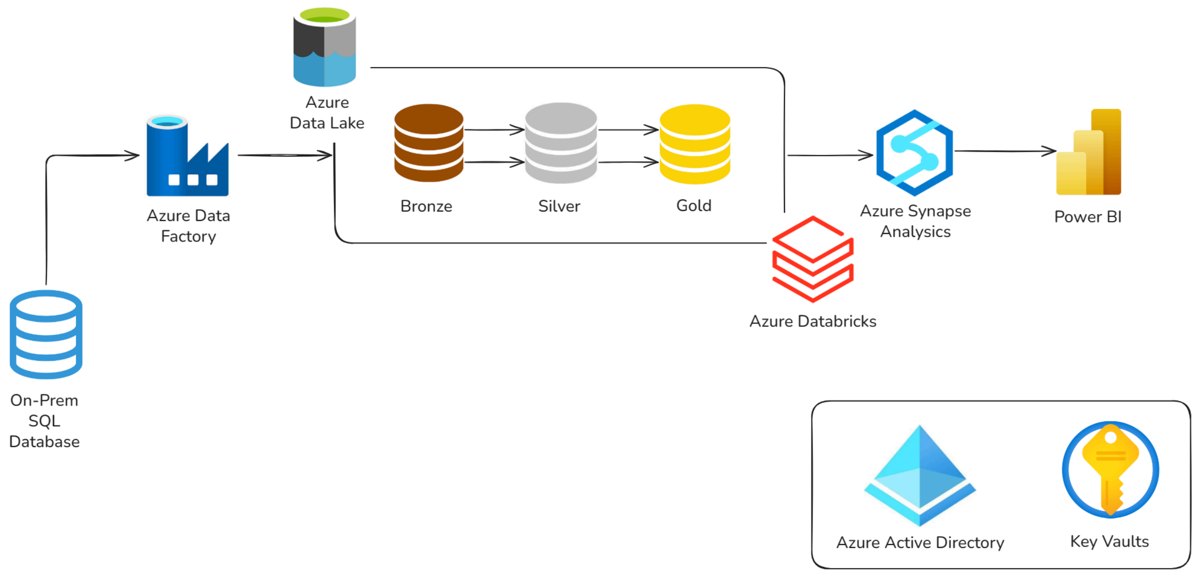


Figure 4: End-to-End Data Pipeline Architecture

4.1 Data Sources

The pipeline starts with a structured raw dataset which is the AdventureWorks sample dataset downloaded from Microsoft website. These datasets contain important operational information including customer records, addresses, sales transactions, products and order details. The data is first imported without modification to preserve the original structure and maintain data lineage.

4.2 Bronze Layer (Data Ingestion)

The Bronze Layer is responsible for ingesting raw data exactly as it arrives from the source systems. Azure Data Factory (ADF) is used to automate this process using pipelines, linked services, datasets and copy activities. ADF enables scheduled and scalable ingestion of multiple datasets while reducing manual intervention through parameterized pipelines and looping mechanisms.

The ingested data is stored in Azure Data Lake Storage Gen2 (ADLS Gen2) under the Bronze container. At this stage, no transformation is applied. This layer acts as the historical backup of all source data.

4.3 Silver Layer (Data Transformation)

The Silver Layer focuses on improving data quality through cleansing, validation and standardization processes. Azure Databricks is used in this stage because it provides distributed data processing using Apache Spark, which is suitable for handling large-scale datasets efficiently.

In this layer, date data is modified into a consistent and more convenient format for later usage. In a real world example, issues such as null values, duplicate records, inconsistent formatting, incorrect data types and invalid relationships shall be resolved. Data from multiple related tables may also be joined and normalized to create a cleaner enterprise view. Standard naming conventions and schema validation are applied to ensure consistency across the system.

The transformed data is then stored back into ADLS Gen2 as Delta tables or parquet files for better performance, reliability, and future scalability.

4.4 Gold Layer (Data Warehouse and Serving)

The Gold Layer contains highly refined and business-ready datasets designed for reporting and decision-making. This stage focuses on creating dimension tables, fact tables, aggregated datasets and KPI-driven structures based on business requirements.

Azure Synapse Analytics is used as the serving and warehousing layer where curated datasets are exposed through SQL views and warehouse tables for high-performance querying. This reduces the need for heavy transformation inside reporting tools and ensures that all users access a single source of truth.

The Gold Layer improves reporting efficiency and supports enterprise data governance by providing standardized and optimized datasets for analytics consumption.

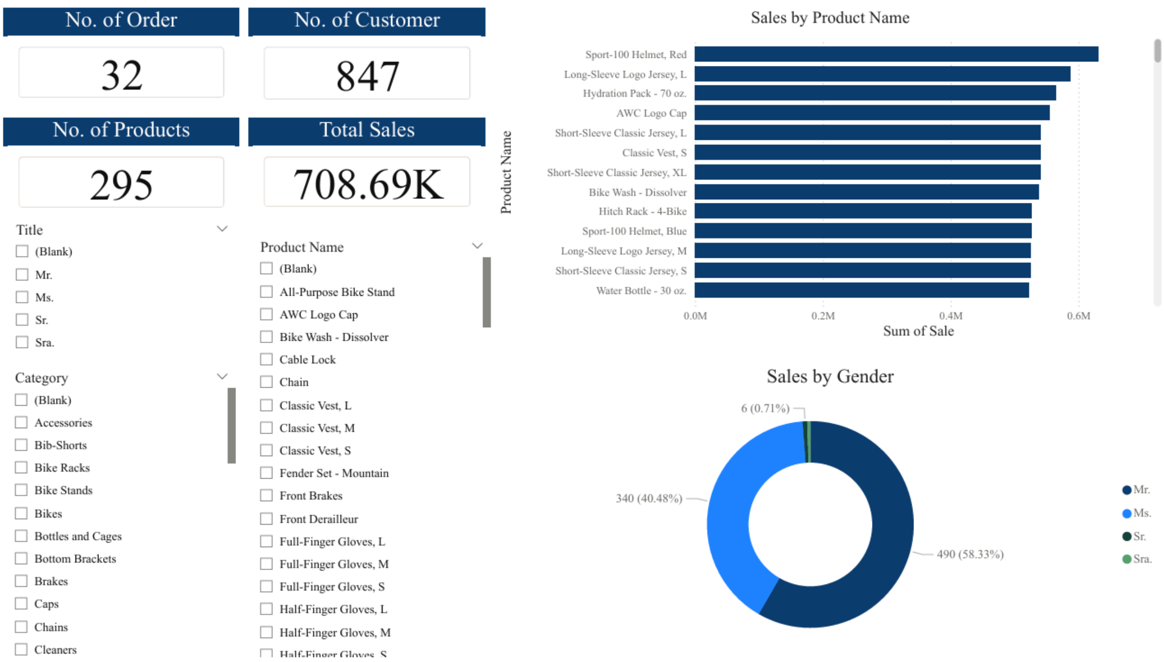
4.5 Analytics and Visualization Layer

The final stage of the architecture is the analytics layer, where Power BI connects to Azure Synapse Analytics to create interactive dashboards and reports. Dashboards are used to monitor KPIs, identify business trends, analyze performance and support strategic decision-making.

Power BI focuses on semantic modeling, visualization and user interaction rather than data preparation. This improves report performance, consistency and long-term maintainability.

The final dashboard serves as the presentation layer of the entire pipeline, transforming technical data processing into meaningful business insights for stakeholders.

5. MS PowerBI Dashboard



6. Reflection

Goe Jie Ying

Through this project, I gained a practical understanding about the structure of a data engineering pipeline in practical terms. Instead of just listening to the theoretical concept in class, step-by-step building the data engineering pipeline helps me better understand how the tools are applied in real-world scenarios. One of the key lessons I learned is the roles of different processes and technologies used at each stage.

During the few days I watched and followed the tutorial video, I realized that the technology evolves rapidly. Although the video was only created 2 years ago, there were noticeable differences in the tools, such as user interfaces and code versions, specifically in Databricks. This created challenges where steps are different with the youtuber and I can't follow every step exactly. To overcome this, I explored alternative solutions and discussed the issues with my teammates to identify possible fixes. It actually took quite a long time to overcome the differences, but it has improved my problem-solving skills and enhanced my understanding of each process in the pipeline. Instead of just copying the content, I can have the chance to think of solutions, which makes my learning journey more meaningful.

Lam Yoke Yu

Through this tutorial, I gained hands-on experience with Microsoft Azure technologies that used to be gigantic to me. Instead of only learning the concepts, I was able to implement them in practice, which gives a basic understanding of how these technologies work together.

I think the most challenging part throughout the project was configuring between services. In particular, when ingesting data from on-premise sources to the cloud. As this was the first time I was using these technologies, there were several settings that had to be configured. While the YouTuber demonstrated the steps seamlessly without issues, I spent a significant amount of time identifying and resolving problems.

Besides, I faced challenges when mounting Databricks to the data lake storage. I spent hours trying to solve the problem. Fortunately, my teammate shared her solution as she faced the same problem. I used to prefer solving problems on my own without seeking help from others, but in this case, the issue was resolved quickly after applying my teammate's code. Moving forward, I will be more open to seeking input from others when facing challenges.

Tan Yi Ya

Through the video tutorial, I gained understanding of how an end-to-end data engineering pipeline is designed and implemented using Microsoft Azure services. Before completing this project, my knowledge of data pipelines was very vague and theoretical, especially regarding the Medallion Architecture. By following the tutorial step by step, I was able to understand the flow from raw data all the way to the business ready dashboard.

Throughout the project, I faced several challenges related to configurations between different services such as importing data into PowerBI from Azure Synapse Analytics SQL, mainly because the file hierarchy was not being set up properly from the previous steps. These problems taught me the importance of troubleshooting, patience and careful system configuration. Overall, I am glad the tutorial existed as this was a valuable experience that truly gives us an overall idea of how the data pipeline is configured and runs automatically.